Introduction to Operating Systems

Computer System Structure

- Computer system can be divided into four components:
  - Hardware – provides basic computing resources
    - CPU, memory, I/O devices
  - Operating system
    - Controls and coordinates use of hardware among various applications and users
  - Application programs –
    - Word processors, compilers, web browsers, database systems, video games
  - Users
    - People, machines, other computers

What is an Operating System?

- A program that acts as an intermediary between a user of a computer and the computer hardware
- Operating system goals:
  - Execute user programs and make solving user problems easier
  - Make the computer system convenient to use
  - Control computer hardware and provide it in an efficient manner

What Operating Systems Do

- Operating Systems whether is a computer or a server provide Hardware control and resource utilization for users and programs achieving optimization , ease of use and high performance .

What Operating Systems Do

- OS is a **resource allocator that**
  - Manages all resources
  - Decides between conflicting requests for efficient and fair resource use
  - Controls execution of programs to prevent errors and improper use of the computer

What Operating Systems Do

User Operating System Interface - CLI

Command Line Interface (CLI) or command interpreter allows direct command entry

- Sometimes implemented in kernel, sometimes by systems program
- Primarily fetches a command from user and executes it

What Operating Systems Do

User Operating System Interface - GUI

- GUI known as graphical user interface is more intuitive visually and all the new users always try to pick this interface instead of CLI

- Many common operations, such as copying and pasting, can be performed by using just the mouse.

What Operating Systems Do

User Operating System Interface - GUI

- Icons and menus take the place of text commands and do not have to be memorized.
- It is also easier to switch between multiple active tasks using a graphical interface. However, advanced operations may only be accessible via the command line.

What Operating Systems Do

- OS manages computer memory, storage , I/O devices and processes.
- While OS interact with application in high language it interact with Hardware with machine language.
- OSs work on various devices from Phones to computers and till servers and mainframes.

Most used OSs

- Microsoft Windows.
- Linux
- Apple IOS.
- Aplle MacOS
- Android.
- Unix

Conclusion

- OS act as a mediator between user and computer hardware .
  - It communicates with other parts of computer such as CPU , memory , storage devices , input /output devices to insure smooth performance and programs execution.
- After OS is loaded into the computer by a boot program it manages all applications and control computer hardware and resources sharing .
- So without OS user would not be able to use apps and hardware .

Types of Operating Systems

- General purpose operating systems.
- Embedded operating systems.
- Mobile operating Systems.
- Real time Operating Systems.

General Purpose Operating Systems

- General purpose operating systems intends to run a multitude of applications on a broad selection of hardware enabling a user to run multiple tasks and applications simultaneously .
- It runs applications from accounting systems and databases to web browsers.
- General purpose operating systems focus on process (thread) and hardware management.

General Purpose Operating Systems

Windows

- The most used OS today.
- Closed Source operating system :
- The software is protected rights and payed license from Microsoft …the software and its update and upgrade can only be provided by Microsoft.

Windows Versions

Windows versions

- Windows XP
- Windows Vista
- Windows 7
- Windows 8
- Windows 10
- …….

General Purpose Operating Systems

- **LINUX**
  - Efficient fast-performing operating systems.
  - Open source operating system .
  - Source code is available for everyone or free .
  - Modification , update and upgrade is done by everyone everywhere.
  - High security , functionality and stability.

Linux Versions

- Linux has many versions due to its open source : (more than a thousand)
  - Ubuntu
  - Fedora
  - Kali
  - Backtrack
  - Red hat

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